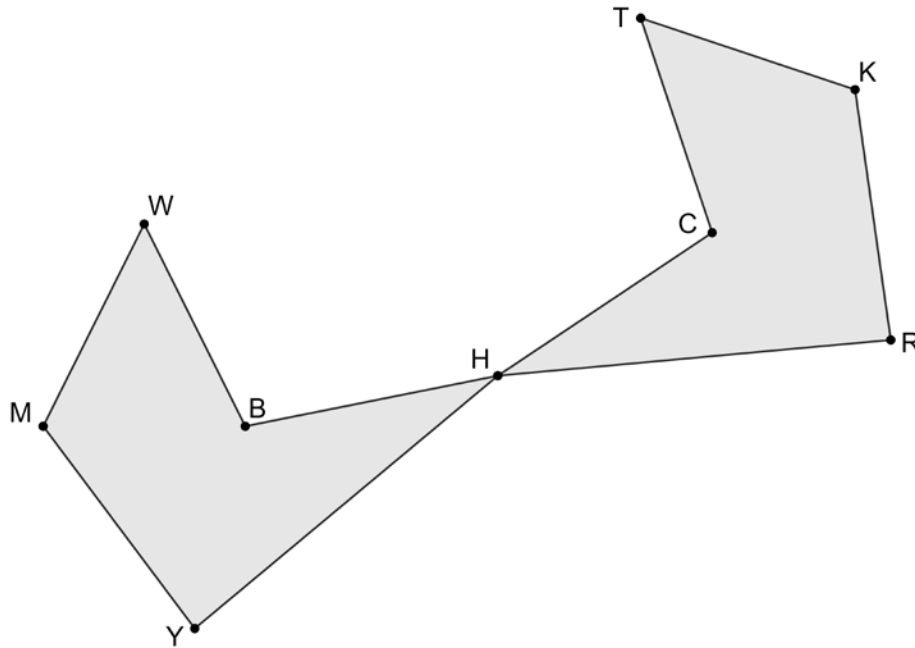


Congruent polygons $WBHYM$ and $TCHRK$ are shown where $\overline{TC} \cong \overline{TK}$.



1. Determine the length of $\overline{TC}$ if $TC = 8x - 11$ and $TK = 3x + 24$.
2. Determine the value of y when the perimeter of $WBHYM$ is 236 units and $HR = 7y - 6$, $KR = 5y - 11$, and $BH = 4y + 3$.
3. What is the length of $\overline{HR}$?
4. What is the length of $\overline{MY}$?
5. What is the length of $\overline{HC}$?

Some of the angle measures of $WBHYM$ and $TCHRK$ are shown.

$$m\angle WMY = (8k - 19)^\circ$$

$$m\angle HRK = (5k + 2)^\circ$$

$$m\angle KTC = (3k + 2)^\circ$$

$$m\angle RHC = (2k - 6)^\circ$$

$$m\angle WBH = 9(m\angle RHC) + 3^\circ$$

$$m\angle RKT = (5k + 32)^\circ$$

6. Determine the value of k .

Determine each angle measure.

7. $m\angle BWM = \underline{\hspace{2cm}}$

8. $m\angle WBH = \underline{\hspace{2cm}}$

9. $m\angle MYH = \underline{\hspace{2cm}}$

10. $m\angle TCH = \underline{\hspace{2cm}}$